Response to reviewers

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Blue-marked and clean manuscripts accompany this major revision. The original 180-run evidence has been recovered and audited. A new 240-trial, 1,080-call matched inference-block study is complete and independently reanalysed from retained records. Independent expert annotation and live O-RAN deployment remain unavailable; the corresponding claims are narrowed.

Associate Editor

> *Overall, the manuscript addresses a timely and relevant problem and presents a promising policy-governed framework for LLM-assisted O-RAN security management. However, significant concerns remain regarding the realism and generality of the evaluation, the interpretation of the LangGraph comparison, and the distinction between semantic correctness and fail-closed execution safety. The implementation and formal verification boundaries should also be clarified more explicitly. These issues require substantial additional validation and analysis before the manuscript can be considered for publication.*

We have clarified implementation and finite-model boundaries, documented all six synthetic scenarios, narrowed accuracy claims to agreement with the author oracle, and added retained-output replay, formal sensitivity checks and descriptive timing from all 180 original reports. We also completed a prospective common-harness 2×2 inference-block study: 240 trials and 1,080 API calls on 16 October 2026. It crosses one/four assessors with absence/presence of the four-role chain, using matched inputs and request contracts, preserved failures and a prespecified paired analysis. The new evidence supports bounded latency contrasts with mixed semantic changes; it does not explain the historical LangGraph ratio or supply independent expert or live O-RAN validation. All responses below identify concrete changes and their limits.

Reviewer 1 — Comment 1: Practical O-RAN integration

> *Clarify the practical O-RAN integration. The paper positions the framework around SMO/Non-RT RIC governance, but the evaluation appears to use synthetic scenarios and simulated actuation rather than real O-RAN telemetry, rApps, A1 policies, or Near-RT RIC/xApp interactions. The authors should more clearly explain what has been implemented versus what remains architectural.*

We have separated the operational architecture from the evaluated prototype. Section IV-A and new Table II distinguish implemented orchestration and policy checks, simulated actuation, the finite formal model, public-testnet experiments and future interface integration. Section VI-D and Table IV identify the actual scenario fields and experimental operator-policy clauses. The live demonstration terminates at SIMULATED_O_RAN_ACTION_LEDGER with production_oran_execution=false and network_side_effects=false. No deployed rApp, standards-conformant A1 delivery, real E2 control or operator telemetry is demonstrated. The SMO/Non-RT RIC placement is therefore an intended management integration context. The abstract, claim boundary and conclusion now use that scope consistently.

Locations: Main IV-A/Table II; VI-D/Table IV; VII-A; VIII.

Reviewer 1 — Comment 2: Benchmark realism

> *Strengthen the benchmark realism. The 180-case frozen benchmark is useful, but it is synthetic. The paper should describe how the scenarios were generated, how representative they are of real O-RAN security operations, and whether they cover realistic ambiguity, conflicting evidence, and noisy telemetry.*

We have documented the construction and limits of all six author-designed configurations in Section VI-D and Table IV. The scenarios cover a compliant O-CU configuration, DRB-integrity policy conflict, legacy TLS, unauthorised xApp access, replayed evidence and missing or unsupported information. P-01–P-04 are experimental operator-policy rules, not normative O-RAN clause numbers. The canonical 180 cells cross six scenarios, three retrieval modes and ten repeats; they are not 180 independent incidents. We verified the exported input-side payloads and retained their hashes. Conflicting policy evidence and uncertainty are represented in these cases, but real noisy telemetry, systematic stale-evidence tests and an independently validated incident distribution are absent. We therefore narrowed the claims and explicitly separated the configured impairment/profile study from operational telemetry validation.

Locations: Main VI-D/Table IV, VI-F, VII-L; Supplement S7.3–S7.4, S10 and S11.1.

Reviewer 1 — Comment 3: Formal verification scope

> *Clarify the formal verification scope. The TLA+/TLC result over 320 reachable states is a useful safety check, but it verifies a finite abstraction, not the full implementation or O-RAN deployment. The paper should more explicitly state which properties are verified and which risks remain outside the model.*

Section V now states the exact six predicates, the seven-field state abstraction and its exclusions. We reran the unchanged model with TLC 2.19: 64 initial states and 320 reachable states, with no invariant violation. Its initialisation excludes a zero commitment. A separate, preserved zero-input probe admits that value and checks 96 initial and 480 reachable states, again without invariant violations. Three deliberate mutations each trigger the expected invariant: removing admissibility, removing the nonzero execution guard, and omitting ordinary approval for PRIVILEGED. Logs, model hashes and a reproducible runner are included. These checks establish bounded safety and sensitivity to those mutations. They do not establish liveness, cryptographic proof validity, code refinement or deployment safety. Duplicate rejection has no history variable in this model and is now confined to the appropriate runtime/contract evidence.

Locations: Main V; Supplement S7.9 and S10.2; evidence/formal/formal_revision_checks.json.

Reviewer 1 — Comment 4: Operational latency

> *Discuss latency implications more deeply. End-to-end decision times of 80.5-109.8 seconds make the framework unsuitable for many operational security responses. The authors correctly state that this is for non-real-time management, but they should define the practical use cases where such latency is acceptable.*

The revised results distinguish reported decision duration from operational deadline compliance. We cite ETSI TS 103 982 V14.0.0 for Near-RT loops around 10 ms–1 s and Non-RT rApp loops above 1 s. The medians are 80.453–109.793 s and the Agentic-RAG P95 is 164.747 s, excluding later human waiting. We restrict intended tasks to deliberative compliance review, scheduled policy-change preparation and remediation planning, subject to a task-specific deadline and evidence-freshness check. We do not claim Near-RT RIC

compatibility or suitability for urgent inline containment. Section VII-J and supplementary Table S15 now analyse all 180 recovered benchmark reports, with 60 timestamp records per mode. The aggregate-to-reflection phase spans the role chain and surrounding processing; its medians are 57.698, 75.637 and 79.608 s for No RAG, RAG and Agentic RAG. Four disjoint intervals sum to each internal report duration. Marginal phase medians do not sum to a median total, and parallel assessor calls are not added to wall time. This is descriptive attribution within the proposed workflow, not a causal framework/component cost. The separate S2 demonstration remains an illustrative supplementary trace. The new common-harness study in VII-K separately reports all-condition decision-ready medians of 4.266, 24.514, 67.514 and 93.144 s and R7 P95 values of 8.069, 50.916, 93.998 and 130.990 s, including failure paths. Its boundary excludes retrieval generation, human waiting and anchoring. These measurements do not establish task deadlines or Near-RT suitability. Expiry and conflict handling remain deployment requirements, rather than tested guarantees.

Locations: Main VII opening, VII-A, VII-J–K/Tables XIII–XV; Supplement S10.3/Table S15 and S11.4.

Reviewer 1 — Comment 5: LangGraph comparison

> *Clarify the comparison with LangGraph. The external LangGraph baseline is helpful, but it is not clear whether the comparison is fully fair. The authors should explain whether LangGraph received the same retrieval inputs, policy constraints, prompts, and approval logic, or whether the proposed workflow benefits from additional governance layers not available to the baseline.*

New Table IX reports the actual parity and differences. The canonical input-side configurations, retrieved passages and reference scoring are shared. The proposed workflow uses four requested provider aliases and a role chain; LangGraph uses one hosted assessor. Their prompts, output budgets, sampling fields, governance, timing boundaries and run dates differ. LangGraph uses a 256-token limit; the proposed OpenAI/Anthropic/Mistral requests use 5,000-token limits and omit sampling fields. The baseline latency surrounds the assessor request, while proposed values cover the returned workflow. LangGraph has no equivalent live review or ledger path. We explicitly qualify the comparison as configured workflows, withdraw framework/component-level causal attribution, and state that coverage definitions differ. The retained-output replay in VII-J remains a separate post-processing ablation. New VII-K adds a genuinely fresh common-harness inference-block comparison. Shared assessor request bodies match within each block; all outputs are newly requested. The common aggregation rule explicitly adapts to one or four configured providers, and a common review-only guard applies to all conditions. Sampling/reasoning defaults and immutable provider snapshots are not inferred. This new study estimates the configured assessor-set and whole-chain contrasts, without relabelling the old comparison as matched or attributing its ratio to n8n/LangGraph overhead.

Locations: Main VII-C/Tables VIII–IX and VII-J–K/Tables XIII–XV; Supplement S6.2, S7.5, S10.1 and S11.

Reviewer 1 — Comment 6: Human approval and accountability

> *Human approval and governance need more detail. Since human approval is central to the framework, the paper should clarify the operator role, approval interface, accountability trail, and how conflicting or delayed human decisions are handled.*

We added two paragraphs in VII-A describing the operator role, recorded review fields, authority boundary and proposed approval view. The archived callback records identifier, claimed role, decision and rationale, linked to the run/report. Those fields do not establish cryptographic authentication. The demonstrated HUMAN approval permits one simulated write; it does not validate rejection, expiry, concurrent decisions, replay, revocation or PRIVILEGED handling. We state that a deployable service must bind the exact action/evidence

version, authenticate authority and keep conflicting or stale decisions blocked. The code audit also found Boolean string coercion in the archived callback. A separate inactive V2 export now requires a real Boolean and nonempty identifiers. Local tests exercise two valid branches and twelve malformed payloads. Historical exports remain unchanged; the new export is not deployed and these tests do not verify gateway authorisation.

Locations: Main VII-A/Fig. 3; Supplement S10.4; inactive L4FinalPaperReviewV2.json and payload tests.

Reviewer 1 — Comment 7: Blockchain and testnet evidence

> *Avoid overclaiming blockchain/testnet evidence. The public-testnet latency results demonstrate anchoring feasibility, but they do not prove security, finality suitability, or production audit reliability. The manuscript already notes some boundaries, but this limitation should be made more prominent in the discussion.*

We made the evidence boundary prominent beside the anchoring results and in the implementation table. Public-testnet measurements demonstrate client-observed first inclusion for the tested bytecode/payload and RPC conditions. They do not isolate consensus effects, measure economic finality or prove production security or audit reliability. A hash can bind available content to a commitment; it does not establish telemetry truth, durable content availability, key security or reviewer authority. We also distinguish the live report's ANCHORED status and local hash chain from an externally verified receipt: the retained live report marks external-chain evidence MISSING. Public-testnet receipts belong to a separate experiment and do not prove anchoring of that live decision. The proof boundary remains explicit: the retained 240-row matrix supports no row-bound successful zero-knowledge verification.

Locations: Main IV-C, VI-F, VII-E–H and Table II; Supplement S7.10 and S10.

Reviewer 1 — Comment 8: Readability and organisation

> *Improve readability and organization. The manuscript and supplementary material are dense. A concise table separating implemented, simulated, formally verified, and future deployment components would greatly improve clarity.*

New Table II gives the requested component-status summary. Table IV maps the six scenarios, and Table IX makes baseline differences visible. The semantic summary, formerly Table IV and now Table VI, contains agreement and coverage only; latency remains in Fig. 4 and safety/declaration interpretation is discussed separately. We revised terminology throughout and added six primary-source related-work entries with a scope-based comparison table. Fig. S1 is now an editable functional map generated from the archived n8n JSON, with a 54-node inventory (45 computational and 9 documentation nodes); a complete editable edge-level diagram is supplied. Fig. S2 is an editable callback map. Supplement S3 now names the source JSON and generator and explains the six functional groups and all 49 exported connections. Their captions distinguish exported structure from observed execution. Public-testnet Table X is split into two panels to restore readable type. Reference years, volumes, pages and DOIs were reconciled with publisher metadata. An acknowledgment discloses AI assistance with revision text, analysis scripts, the matched-study harness and reconstructed workflow figures, while identifying the author-executed API study as the source of its retained responses. The marked main manuscript and supplement compile, and clean versions are supplied.

Locations: Main II, Tables I–II, IV, VI, IX–X and XIV–XV; Supplement S3/Figs. S1–S2 and S10–S11; editable sources.

Reviewer 2 — Comment 1: Oracle-relative accuracy and generality

> *The 180-cell evaluation is limited to six synthetic scenarios, and the oracle lacks independent expert validation. Therefore, the reported accuracy should be presented as oracle-relative, not as evidence of general O-RAN correctness; broader validation or narrower claims are needed.*

We adopted the narrower-claim option. The abstract, methodology, results, Fig. 4, Table VI and conclusion describe agreement with the frozen experimental oracle, rather than general O-RAN correctness. We explicitly state that six distinct author-designed scenarios have no completed independent expert validation. Ten repeats describe stability within scenario-mode conditions. All 180 reference actions require non-automatic handling (120 HUMAN, 60 NEVER), so the benchmark does not test useful autonomous execution or the full four-class distribution. We preserved the frozen oracle and verified its exported inputs, manifest and baseline joins. The existing human-validation protocol is identified as future work, including the need to amend and freeze its analysis plan before new labels are collected: action classes are nominal, so the archived linear-weighted statistic lacks a justified ordering. We have now recovered all 180 original report packages and verified their manifests (3,240 bound files plus 180 manifests). Every canonical report hash joins the previously released frozen reference. Raw reanalysis reproduces the decision/action counts, NEVER recall, coverage, whole-scenario bootstrap intervals and indexed latency statistics. The original evidence and deterministic audit script are included in the revision archive. These integrity checks do not independently authenticate provider services or validate the synthetic oracle. The new 240-trial study reuses the same six scenarios; additional model calls do not increase scenario diversity or constitute independent expert validation. Its results are reported separately and do not replace the original 180-cell results.

Locations: Main abstract, VI-D/Table IV, VII-B/Table VI/Fig. 4, VII-K–L and VIII; Supplement S7.3–S7.4, S10.1 and S11.

Reviewer 2 — Comment 2: Ablation and the latency difference

> *The LangGraph comparison changes multiple architectural components simultaneously, preventing attribution of the reported gains to any single mechanism. Given the substantial latency difference (~109.8 s vs. 1.5-1.8 s), an ablation study or more qualified interpretation is needed.*

We have qualified the historical comparison and added a new prospective 2×2 common-harness inference-block study, rather than inferring a mechanism from the old ratio. On 16 October 2026 we ran all 60 frozen Agentic-RAG cells under A: one assessor/no roles; B: four assessors/no roles; C: one assessor/four roles; D: four assessors/four roles. There are 240 fresh trials and 1,080 API calls. The protocol was frozen before the full run after two separate pilots. Shared assessor bodies, evidence, aliases and output caps match within blocks; outputs are not replayed. The configured-count aggregation adaptation, four-role continuation rule and common review-only guard are explicit. All conditions use 5,000-token assessor and 8,192-token role caps, one attempt and a 180 s request timeout. The timer includes prompt construction, inference, common processing and intermediate node-file persistence; retrieval generation, human waiting, anchoring and final index/manifest writes are excluded.

Prespecified paired means use 59 blocks with all planned HTTP responses 200; block 7 is excluded for a Mistral read timeout. HTTP completion includes refusals and is not successful inference. These complete-case contrasts condition on transport completion; prespecification does not eliminate possible selection bias relative to all attempted blocks. Adding the assessor set costs 23.622 s without roles and 27.017 s with roles; adding the whole role chain costs 63.052 s with one assessor and 66.446 s with four. The interaction is 3.395 s. Main Table XV reports descriptive 95% percentile intervals from 20,000 whole-scenario bootstrap draws.

With only six designed clusters, we make no significance, reliable nominal-coverage or general O-RAN claim. Per-condition median/P95 descriptions retain all 60 trials, including failures.

The semantic changes are mixed. A/B/C/D decision agreement is 52/54/52/59 out of 60, pre-review action agreement 55/51/55/48, and post-guard agreement 59/58/59/58. NEVER recall is 19/18/19/18 out of 20. C does not improve decision agreement over A, and D has lower NEVER recall and pre-review action agreement than A. The common guard imposes non-execution and can increase final action agreement without semantic understanding. The full node audit retains 33 Anthropic refusals, nine DeepSeek length terminations, six reference-check failures and one timeout. Four truncated DeepSeek objects were accepted through the frozen reasoning-channel fallback; this is disclosed explicitly. Supplement S11 reports post hoc validity/completion sensitivities as outcome-selected descriptions, not replacements for the prespecified analysis. Neither pilot is pooled and no failure is replaced or repaired.

Table XIII's retained-output replay and supplementary Table S15's all-180 historical phase timing remain separate evidence. The historical 109.793/1.843 ≈ 59.6 ratio still compares unequal workflows, budgets, dates and timing boundaries. The new experiment does not retrospectively decompose that gap, estimate framework overhead, isolate individual roles or separate provider identity from assessor count. It provides the requested short ablation within an explicitly bounded, reproducible estimand.

Locations: Main VII-C/Table IX and VII-J–K/Tables XIII–XV; Supplement S10.1, S10.3/Table S15 and S11/Tables S16–S18; matched_inference_audit.json.

Reviewer 2 — Comment 3: Semantic correctness and execution safety

> *Zero unsafe executions primarily demonstrate effective fail-closed governance, not semantic correctness. The manuscript should clearly distinguish semantic accuracy, policy behavior, and execution safety.*

The revision separates three outcomes: oracle-relative decision/action agreement, policy routing/coverage, and observed execution at an explicitly identified target. The raw-artifact/code audit resolves the local Qwen wording: the legacy unsafe_execution field means oracle_action_class == NEVER and final_prediction.execute == true. Neither baseline runner invokes an executor. Qwen’s nine flags therefore denote erroneous declarations, not observed RAN or simulated executions; the broader count is 38 declarations on reference non-automatic cases. LangGraph has zero such flags on 180 cells. Original records are retained unchanged and the corrected interpretation is recorded in a derived report. The proposed benchmark’s disabled automatic execution and the separate single approved simulated write have distinct scopes. The replay ablation shows that suppressing declarations can leave semantic errors unchanged. The revised results and conclusion distinguish disabled automatic execution in the proposed benchmark from zero execution declarations in LangGraph; neither is evidence of semantic correctness. Supplementary Table S14 now labels the oracle-NEVER output flags as declarations and states that no executor is invoked. We also corrected the proposed Agentic-RAG NEVER-class recall from 100% to 95% (19/20), recomputed from original final-action records and cross-checked against retained histograms. That recall is independent of the execute flag; the derivation and regression check are included. The new matched experiment likewise reports decision, pre-review action, post-guard action, NEVER recall, structural/reference validity and transport completion separately. Its common review-only guard makes zero execution inevitable, so we do not present that as an additional safety achievement.

Locations: Main VI-D, VII-B–D, VII-J–K and VIII; Tables VI–VIII and XIII–XV; Supplement S10.1 and S11.5; retained and matched evidence audits.